(2a)

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Taken	CS1, CS0	NS1, NS0	Predict.
0	0 0	0 1	1
0	0 1	1 1	1
0	1 0	1 1	0
0	1 1	1 1	0
1	0 0	0 0	1
1	0 1	0 0	1
1	1 0	0 0	0
1	1 1	1 0	0

2B: FSM controller diagram

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diagram

Taken	Cs1	Cs0	Predict
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

← not affected by Taken or Cs0.

Predict = (Cs1)'

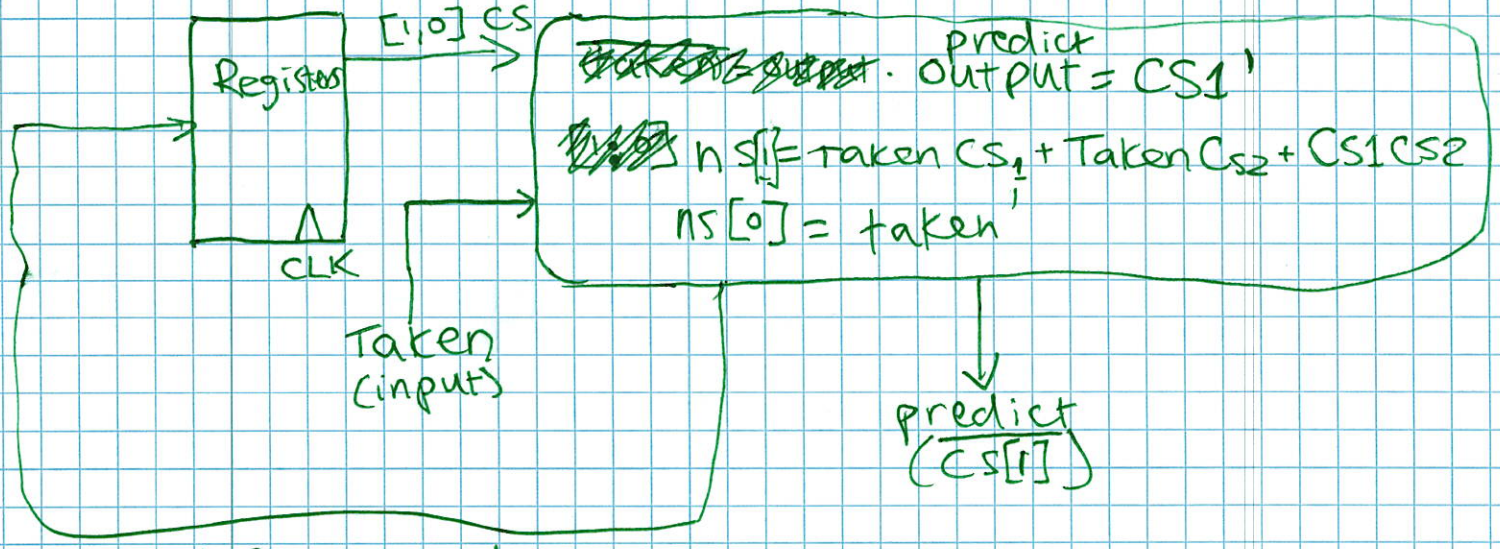
Taken	Cs1	Cs0	Predict	Ns1	Ns2
0	0	0	1	0	1
0	0	1	0	1	1
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	1	0	0
1	1	1	0	1	0

Ns1 ↑ Taken ≠ (CS1 OR CS2)

Ns2 ↑ taken'

*

Taken·Cs1 + Taken·Cs2 + Taken' Cs1 Cs2 · { Taken ≠ (CS1 OR CS2) / (Taken' or and (cs1 and Cs2).



NS[0] = taken'

NS[1] = Taken · CS1 + Taken CS2 + CS1 CS2.

* TC1 + TC2 + T̄C1C2.
= TC1 + TC1C2 + TC2 + T̄C1C2
= TC1 + TC2 + C1C2(T + T̄)
= TC1 + TC2 + C1C2.

used info from:
cs.cornell.edu/courses/cs3410/2018fa/schedule/slides/07-fsm.pdf .

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(2c) Answers typed & inserted into scan of PDF.